ELEC5208 Active Distribution Systems

Group Assignment

TASK II: Home Energy Management System

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1 Introduction

The increasing penetration of rooftop photovoltaic (PV) systems and residential battery storage is changing the operation of low-voltage distribution networks. While rooftop PV can reduce household electricity costs and improve renewable-energy utilisation, high PV penetration may also introduce operational issues such as reverse power flow, voltage rise, and voltage unbalance.

In Task I, a full-year quasi-static time-series (QSTS) simulation of a low-voltage residential feeder was developed in OpenDSS using real smart-meter demand data. In Task II, this network is extended by integrating rooftop PV systems, home batteries, and a rule-based Home Energy Management System (HEMS).

The rule-based HEMS coordinates PV generation, battery charging/discharging, household demand, and grid interaction using predefined operational rules. Unlike optimisation-based approaches, the controller follows simple deterministic priorities, including:

- PV supplying household demand first,
- charging the battery using surplus PV generation,
- exporting remaining surplus to the grid,
- discharging the battery during peak-price periods.

A 24-hour HEMS simulation was first implemented in Julia to analyse battery operation under different battery capacities and tariff conditions, including the Solar Share Offer (SSO) scheme with free midday electricity. The HEMS was then integrated into the OpenDSS feeder model to perform full-year simulations for several case studies:

- rooftop PV only,
- rooftop PV with 10 kWh batteries,
- rooftop PV with 20 kWh batteries,
- SSO-enhanced scenarios.

The simulations were used to investigate the impacts of PV and battery integration on feeder active power, reverse power flow, voltage behaviour, and voltage unbalance. In addition, the effectiveness and limitations of the rule-based HEMS strategy were evaluated through comparison of the different case studies.

2 Rule-Based HEMS Development

2.1 HEMS Assumptions and Operational Rules

The Home Energy Management System (HEMS) developed in this study coordinates rooftop PV generation, residential demand, battery storage, and grid interaction at each simulation time step. A deterministic rule-based control strategy was implemented instead of an optimisation-based controller. This approach was selected because it is computationally simple, easy to interpret, and consistent with the assignment requirement for a rule-based HEMS.

Several operational assumptions were adopted in the model. The battery charging/discharging efficiency was assumed to be 90%, while the maximum charging and discharging power was limited to 5 kW according to the assignment specification. Two battery capacities were investigated: 10 kWh and 20 kWh. The battery state-of-charge (SOC) was constrained within its capacity limits to avoid overcharging and overdischarging.

The controller follows a priority-based operational logic. Rooftop PV generation first supplies local household demand. If surplus PV generation exists, the battery is charged subject to SOC and power constraints. Any remaining surplus power is exported to the upstream grid. Conversely, when PV generation is insufficient to satisfy the household load, the battery is allowed to discharge during peak-price periods to reduce grid imports. Any remaining deficit is then supplied by the grid.

Mutual exclusivity was enforced throughout the simulation. The battery cannot charge and discharge simultaneously, and grid import/export cannot occur at the same time. This ensures physically feasible battery operation and prevents unrealistic power-flow behaviour.

In addition to the standard Time-of-Use (TOU) tariff, the Solar Share Offer (SSO) scenario was also investigated. Under the SSO scheme, electricity is assumed to be free between 11:00 and 14:00. During this period, additional battery charging from the grid is permitted in order to evaluate how tariff incentives influence battery operation and feeder behaviour.

Variable	Value	Unit
Battery Size	10	kWh
Battery Charging/Discharging Rate	5	kW
Battery Efficiency	90	%

Figure 1: Battery Parameters

Tariff	Price	Duration
Shoulder (Day)	0.25	07:00 to 13:59
Peak	0.35	14:00 to 19:59
Shoulder (Night)	0.25	20:00 to 21:59
Off-peak	0.15	22:00 to 06:59
Feed-in (Export)	0.08	–

Figure 2: Electricity Prices and Tariffs($/kWh)

2.2 Julia Implementation and 24-hour HEMS Simulation

The rule-based HEMS was implemented in Julia to simulate the interaction between rooftop PV generation, household demand, battery storage, and grid power exchange over a 24-hour period. The simulation was performed using hourly aggregated smart-meter data derived from the original 5-minute SolarAnalytics dataset.

A representative residential load profile was selected for the study. For each hour, household demand and PV generation were aggregated from the original high-resolution dataset to simplify the battery dispatch analysis and improve result interpretability.

Four scenarios were investigated, covering both 10 kWh and 20 kWh battery capacities under the standard TOU tariff and the SSO tariff:

10 kWh battery under the standard TOU tariff,

20 kWh battery under the standard TOU tariff,

10 kWh battery under the SSO tariff,

20 kWh battery under the SSO tariff.

For each scenario, the HEMS controller calculated:

battery charging/discharging power,

grid import/export power,

PV utilisation,

battery state-of-charge (SOC),

daily electricity cost.

The results show that rooftop PV significantly reduces daytime grid imports, while the battery further shifts energy usage from low-price periods to high-price periods.

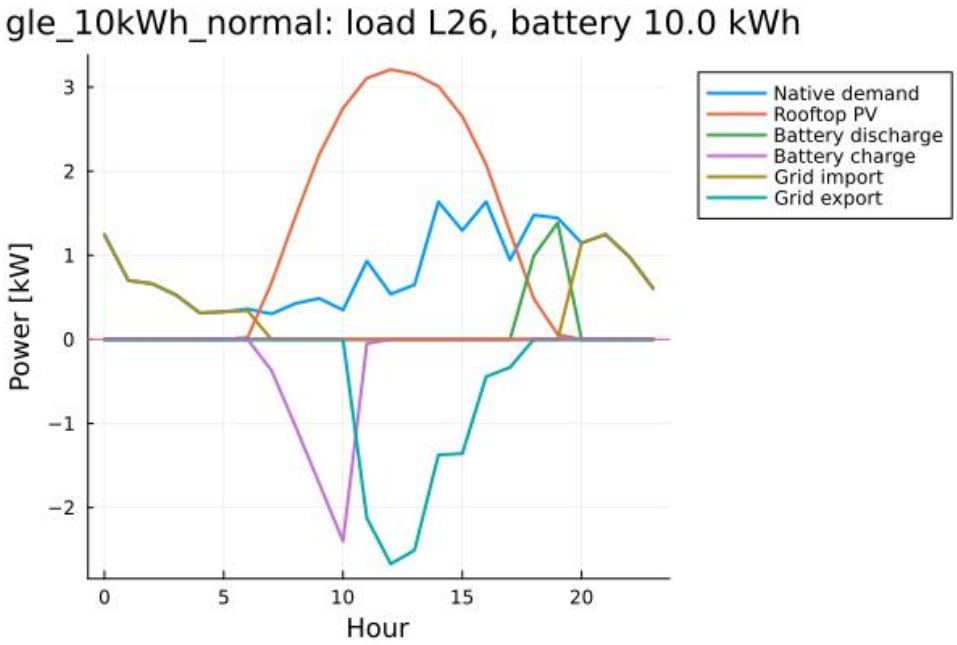


Figure 3: 24-hour Dispatch Result: 10 kWh Battery

Compared with the 10 kWh battery, the 20 kWh battery absorbs more surplus PV energy and reduces reverse power flow more effectively. However, the improvement is limited by the fixed 5 kW battery charging/discharging power constraint.

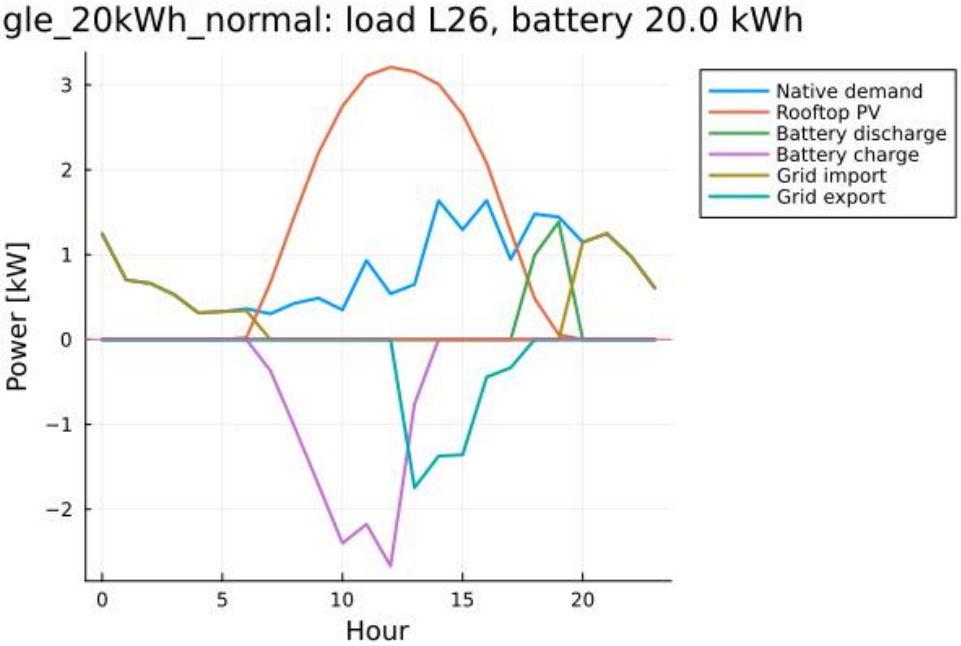


Figure 4: 24-hour Dispatch Result: 20 kWh Battery

Under the SSO scenario, additional battery charging occurs during the free-electricity period. For the 10 kWh battery, the impact of SSO is relatively limited because the battery is already

close to fully charged by midday PV surplus. In contrast, the 20 kWh battery retains more available storage capacity and therefore exhibits more noticeable changes in charging behaviour and feeder power exchange.



Figure 5: 24-hour Dispatch Result under SSO

Overall, the 24-hour simulation demonstrates that battery capacity and tariff structure both strongly influence HEMS operation and feeder power behaviour.

Scenario	Daily Cost (AUD)
10 kWh battery	0.590811
20 kWh battery	1.035255
10 kWh battery + SSO	0.590811
20 kWh battery + SSO	0.809371

Table 1: Daily Cost Comparison

The 20 kWh battery does not necessarily produce the lowest daily operating cost under the standard TOU tariff. Although the larger battery can absorb more PV energy, the charging/discharging power remains limited to 5 kW, reducing the economic benefit of the increased storage capacity.

3 Full-Year OpenDSS Case Studies

3.1 Case 2: Rooftop PV Only

In Case 2, rooftop PV systems were integrated into 15 selected residential loads across the low-voltage feeder without including battery storage. The purpose of this case study was to investigate the impact of high PV penetration on feeder power flow and voltage behaviour under full-year operating conditions.

The same quasi-static time-series (QSTS) simulation framework developed in Task I was used in OpenDSS. Residential load profiles and rooftop PV generation profiles were simulated using 5-minute time resolution over an entire year, resulting in 105120 simulation steps.

Compared with the Task I base case without PV, significant changes in feeder operation were observed after rooftop PV integration. During high solar-generation periods, feeder active power became negative, indicating the occurrence of reverse power flow. This behaviour occurs when total rooftop PV generation exceeds local residential demand and excess power is exported back to the upstream grid.

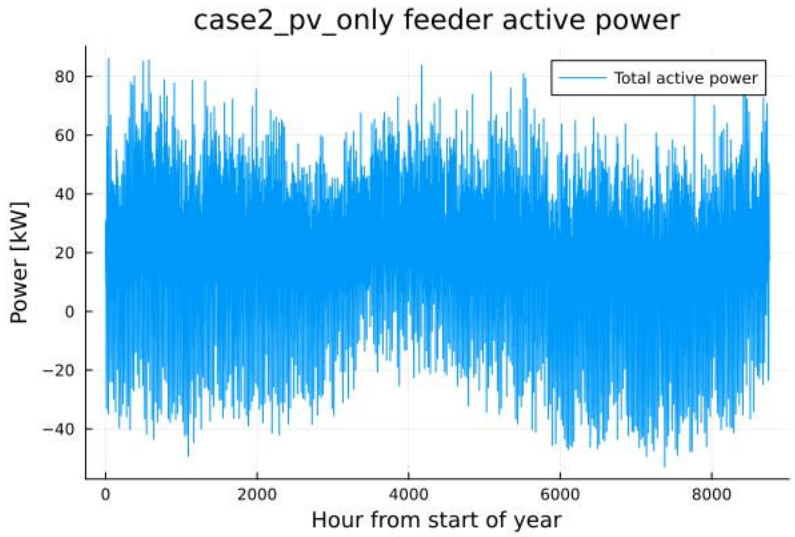


Figure 6: Full-year Feeder Active Power — Case 2 PV Only

The voltage profile also changed noticeably after PV integration. Daytime voltages increased across the feeder due to local active-power injection from rooftop PV systems. Although the maximum feeder voltage increased compared with the Task I base case, the voltage remained within the acceptable operating range of 0.9–1.1 p.u. throughout the full-year simulation.

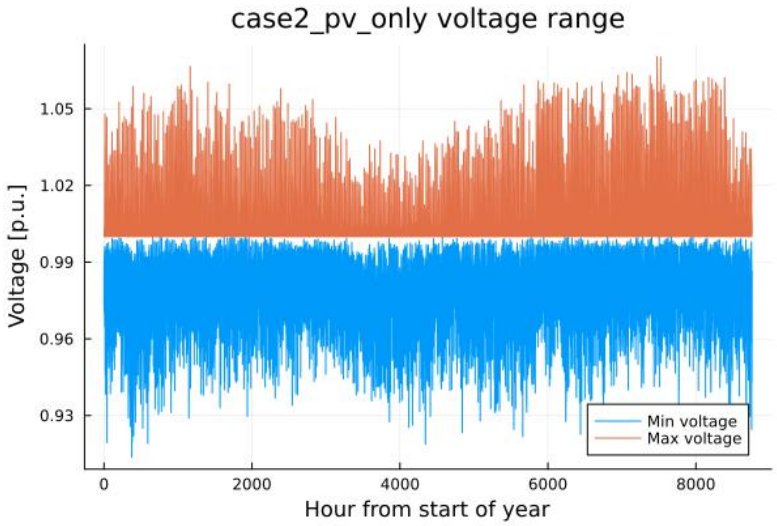


Figure 7: Full-year Voltage Range — Case 2 PV Only

Overall, Case 2 demonstrates that high rooftop PV penetration can significantly alter feeder power-flow characteristics and voltage behaviour, particularly during midday periods with high solar generation. The appearance of reverse power flow and voltage rise confirms that rooftop PV integration significantly changes feeder operating conditions compared with the original Task I network.

Metric	Value
Minimum voltage (p.u.)	0.913709
Maximum voltage (p.u.)	1.070492
Maximum feeder import (kW)	86.15559
Maximum feeder export (kW)	52.90503
Annual feeder import (kWh)	152933.7
Annual feeder export (kWh)	24087.22
Reverse power flow observed	Yes
Reverse power flow duration (hours)	1586
Voltage limit violation	No

Table 2: Case 2 Voltage and Power Flow Summary

3.2 Case 3: Rooftop PV with 10 kWh Batteries

In Case 3, home battery systems were integrated together with rooftop PV generation to investigate the impact of distributed energy storage on feeder operation. A total of 10 residential loads equipped with rooftop PV were selected to include 10 kWh battery systems controlled by the rule-based HEMS developed in Section 2.

The same full-year quasi-static time-series (QSTS) simulation framework was used in OpenDSS with 5-minute time resolution. Compared with Case 2, the addition of battery storage noticeably changed both feeder power flow and voltage behaviour.

During periods of high PV generation, the batteries absorbed part of the surplus rooftop PV energy through local charging. As a result, the magnitude of reverse power flow was reduced compared with the PV-only case. This indicates that battery storage improves local PV self-consumption and reduces the amount of excess power exported to the upstream grid.

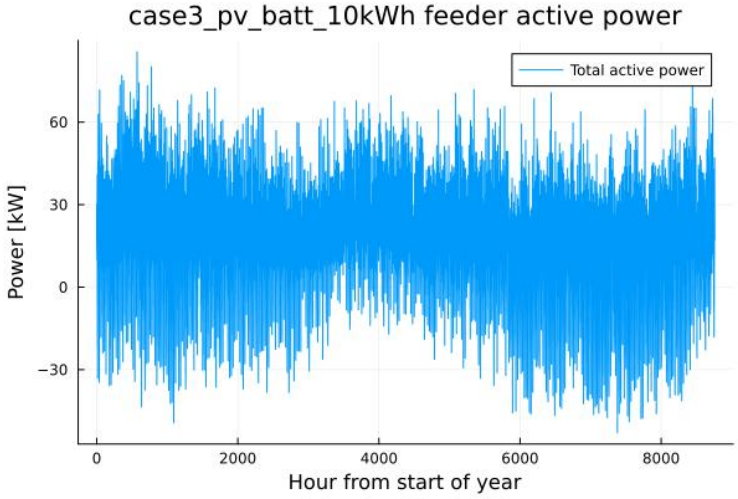


Figure 8: Full-year Feeder Active Power — Case 3 (10 kWh Batteries)

The feeder voltage profile also showed improvement after battery integration. Since battery charging reduced midday feeder export, the voltage rise caused by rooftop PV generation became less severe. Although voltage rise was still present during high-generation periods, the overall voltage variation across the feeder was reduced compared with Case 2.

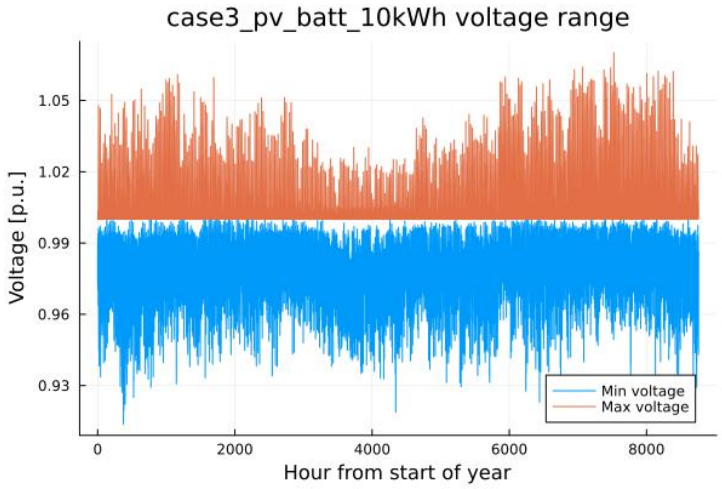


Figure 9: Full-year Voltage Range — Case 3 (10 kWh Batteries)

The minimum feeder voltage remained similar to the PV-only case. This is mainly because the battery discharge strategy was restricted to peak-price periods and the battery penetration level remained relatively limited compared with the total feeder demand.

Overall, the results demonstrate that distributed battery storage can effectively mitigate reverse power flow and reduce PV-induced voltage rise in residential low-voltage feeders. Compared with Case 2, the addition of 10 kWh batteries reduced both reverse power flow magnitude and daytime voltage rise, demonstrating the effectiveness of distributed battery charging in absorbing local PV surplus energy.

Metric	Value
Minimum voltage (p.u.)	0.913725
Maximum voltage (p.u.)	1.070175
Maximum feeder import (kW)	85.57210
Maximum feeder export (kW)	52.90507
Annual feeder import (kWh)	149817.5
Annual feeder export (kWh)	18817.55
Reverse power flow duration (hours)	1319.917
Voltage limit violation	No
Maximum PVUR (%)	6.033553
Maximum VUF (%)	3.517228

Table 3: Case 3 (10 kWh Battery) Summary

Although annual feeder export energy and reverse power flow duration decreased after battery integration, the maximum instantaneous feeder export remained similar to the PV-only case. This is mainly because the battery charging power was limited to 5 kW, restricting the ability of the batteries to absorb short-duration PV peaks. In addition, the maximum PVUR and VUF values indicate that battery integration did not completely remove phase unbalance, because the batteries were installed only at selected single-phase customer locations.

3.3 Case 3: Rooftop PV with 20 kWh Batteries

To further investigate the influence of battery capacity on feeder operation, the battery size was increased from 10 kWh to 20 kWh while maintaining the same 5 kW charging/discharging power limit. The same 10 residential loads equipped with rooftop PV systems were selected for battery integration.

Compared with the 10 kWh battery case, the larger battery capacity allowed more surplus PV energy to be stored locally during high solar-generation periods. Consequently, annual feeder

export energy and reverse power flow duration were further reduced. This indicates that increasing battery capacity improves local PV self-consumption and reduces dependence on the upstream grid.

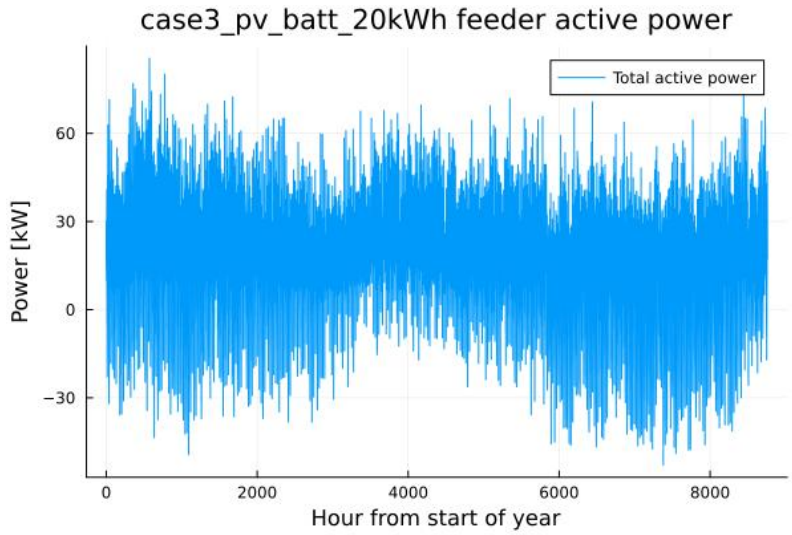


Figure 10: Full-year Feeder Active Power — Case 3 (20 kWh Batteries)

The feeder voltage profile also showed slight improvement compared with the 10 kWh battery case. Since the larger battery absorbed additional midday PV surplus, the severity of daytime voltage rise was further mitigated. However, the overall improvement remained relatively moderate.

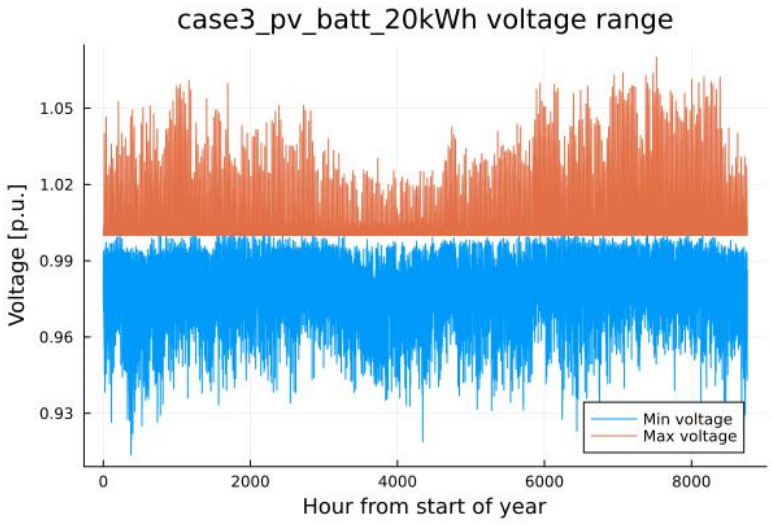


Figure 11: Full-year Voltage Range — Case 3 (20 kWh Batteries)

This behaviour is mainly caused by the fixed 5 kW charging/discharging power constraint. Although the battery energy capacity increased significantly, the battery could still only charge

or discharge at a maximum rate of 5 kW. As a result, the battery was unable to absorb all short-duration PV peaks during periods of very high solar generation.

Overall, the results demonstrate that increasing battery energy capacity alone does not completely eliminate reverse power flow or voltage rise. Battery power rating also plays a critical role in determining the effectiveness of distributed energy storage systems.

Metric	Value
Minimum voltage (p.u.)	0.913725
Maximum voltage (p.u.)	1.070175
Maximum feeder import (kW)	85.57211
Maximum feeder export (kW)	52.90512
Annual feeder import (kWh)	149546.9
Annual feeder export (kWh)	18389.54
Reverse power flow duration (hours)	1301.333
Maximum PVUR (%)	6.033557
Maximum VUF (%)	3.517231
Voltage limit violation	No

Table 4: Case 3 (20 kWh Battery) Summary

Although the 20 kWh battery provided additional reduction in annual feeder export energy, the improvement compared with the 10 kWh battery case was not proportional to the increase in storage capacity. This highlights the importance of battery charging/discharging power limits in residential energy-storage applications.

3.4 Solar Share Offer (SSO) Analysis

To investigate the influence of electricity pricing on HEMS operation, additional simulations were performed under the Solar Share Offer (SSO) scheme. In this scenario, electricity was assumed to be free between 11:00 and 14:00, allowing additional battery charging during midday periods.

For the PV-only case, the introduction of SSO produced almost no noticeable change in feeder power flow or voltage behaviour. Since no battery storage was included in Case 2, modifying the electricity tariff did not significantly alter feeder operation. The network behaviour remained primarily determined by native demand and rooftop PV generation profiles.

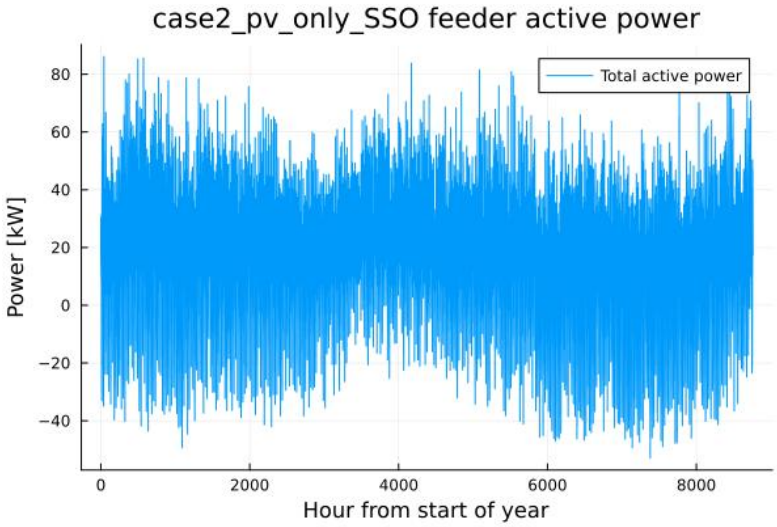


Figure 12: Full-year Feeder Active Power — Case 2 PV Only with SSO

In contrast, the impact of SSO became more significant after battery integration. Under the 20 kWh battery scenario, additional battery charging occurred during the free-electricity period, increasing local energy storage and reducing evening grid imports. As a result, reverse power flow during midday periods was further reduced compared with the standard TOU tariff case.

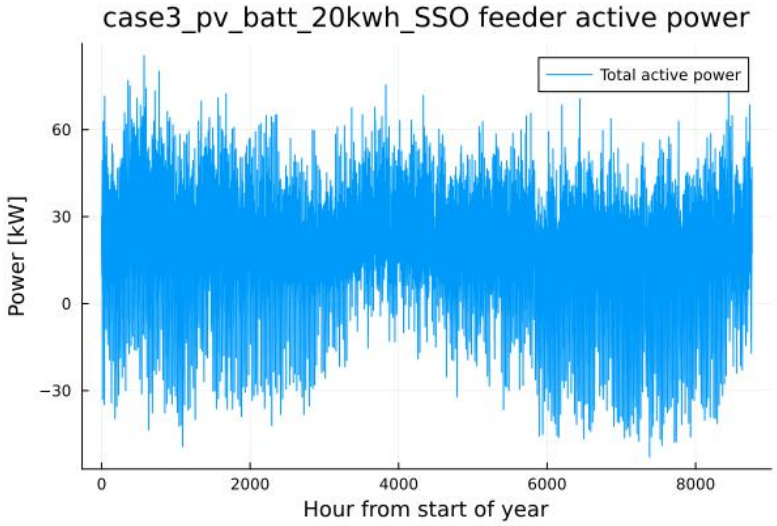


Figure 13: Full-year Feeder Active Power — Case 3 (20 kWh Batteries) with SSO

The SSO scenario reduced annual feeder export energy and reverse power flow duration, especially when batteries were included. However, the voltage profile did not improve uniformly. The maximum voltage remained similar to the non-SSO cases, and the worst-case minimum voltage in the 20 kWh SSO case became slightly lower. This indicates that tariff-driven battery charging changes feeder power-flow behaviour, but it does not necessarily optimise feeder voltage performance.

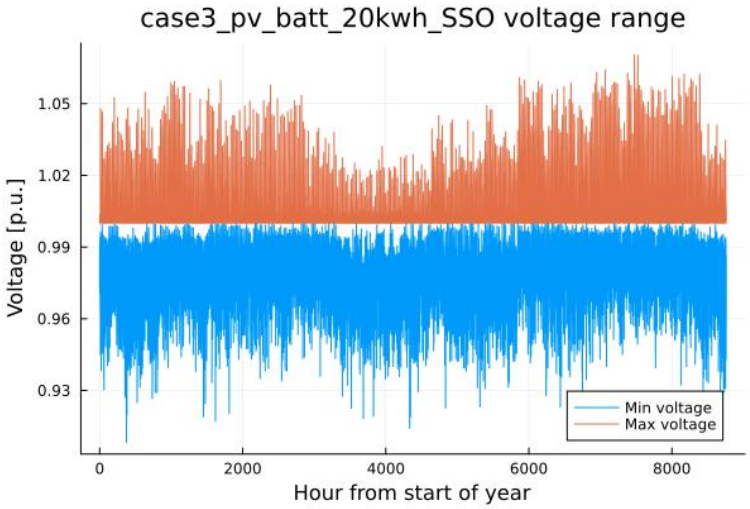


Figure 14: Full-year Voltage Range — Case 3 (20 kWh Batteries) with SSO

However, the SSO results also revealed several limitations of the rule-based controller. During free-electricity periods, the controller allowed relatively aggressive battery charging from the grid. Although this behaviour reduced customer electricity cost, it also introduced temporary increases in feeder import power. This demonstrates that rule-based HEMS strategies may improve local economic performance without necessarily achieving globally optimal feeder operation.

Overall, the SSO analysis demonstrates that electricity tariff structures can strongly influence battery dispatch behaviour and feeder power-flow characteristics.

Metric	PV Only + SSO	20 kWh Battery + SSO
Minimum voltage (p.u.)	0.913709	0.908251
Maximum voltage (p.u.)	1.070492	1.070492
Annual feeder export (kWh)	24087.22	18172.39
Reverse power flow duration (hours)	1586	1243.917
Maximum PVUR (%)	6.467141	7.17101
Maximum VUF (%)	3.388114	3.581458

Table 5: SSO Case Summary

The SSO scenario highlights both the effectiveness and limitations of the rule-based HEMS strategy. Compared with the PV-only SSO case, the 20 kWh battery with SSO reduced annual feeder export energy and reverse power flow duration. However, the worst-case PVUR and VUF increased under the battery SSO case. This suggests that free midday electricity can encourage additional battery charging and reduce reverse power flow, but the resulting single-phase charging behaviour may worsen the worst-case voltage unbalance.

4 Voltage Unbalance Analysis

4.1 PVUR Analysis

To further evaluate feeder performance under high rooftop PV penetration, voltage unbalance was analysed using the Phase Voltage Unbalance Rate (PVUR). Since the residential feeder consists primarily of unevenly distributed single-phase loads and rooftop PV systems, voltage imbalance naturally occurs between the three phases.

The PVUR analysis was performed using the full-year quasi-static time-series (QSTS) simulation results. At each 5-minute time step, the maximum feeder PVUR value was calculated across all buses in the network.

The results indicate that PVUR varied continuously throughout the year depending on feeder loading conditions and rooftop PV generation levels. Most PVUR values remained relatively low during normal operating conditions; however, several higher PVUR events were observed during periods of highly uneven phase loading and strong rooftop PV generation.

The maximum PVUR value reached approximately 6%, indicating noticeable voltage imbalance under certain operating conditions. These higher imbalance levels primarily occurred near feeder-end buses where single-phase PV and residential loads were unevenly distributed.

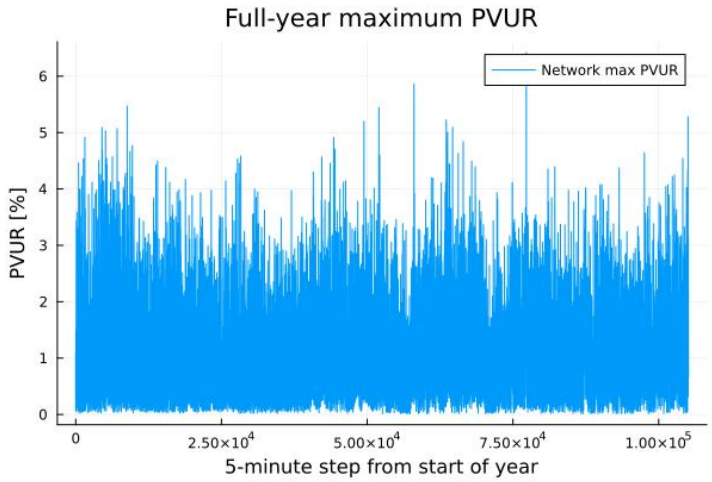


Figure 15: Full-year Maximum PVUR

Across the Task II case studies, the maximum PVUR ranged from approximately 6.03% to 7.17%. The PV-only case produced a maximum PVUR of about 6.47%, while the standard battery cases reduced the maximum PVUR to about 6.03%. However, under the SSO battery scenarios, the maximum PVUR increased to about 7.17%. This indicates that battery integration can reduce

voltage unbalance under standard operation, but tariff-driven battery charging under SSO may shift phase-specific loading and worsen the worst-case PVUR. The highest PVUR was observed at bus b6a, which is electrically remote and therefore more sensitive to phase imbalance.

Metric	Value
Maximum PVUR range across Task II cases	6.033% – 7.171%
Lowest maximum PVUR	6.033% in Case 3 with 10 kWh / 20 kWh battery under standard TOU
Highest maximum PVUR	7.171% in Case 3 with SSO
Bus with highest PVUR	b6a
PVUR variation observed	Yes
Severity	Moderate

Table 6: PVUR Summary

4.2 VUF Analysis

In addition to PVUR, voltage unbalance was also evaluated using the Voltage Unbalance Factor (VUF), which is based on symmetrical component analysis. VUF measures the ratio between negative-sequence voltage and positive-sequence voltage, providing a more rigorous representation of three-phase voltage imbalance.

The VUF analysis was performed for the full-year simulation using the same 5-minute time resolution. At each simulation step, the maximum VUF value across all feeder buses was recorded.

The results show that VUF varied continuously throughout the year due to the time-varying behaviour of residential single-phase loads and rooftop PV generation. Most VUF values remained relatively low during normal operating conditions; however, several higher imbalance events were observed during periods of highly uneven phase loading.

The maximum VUF value reached approximately 3%, indicating noticeable negative-sequence voltage components under certain operating conditions. Similar to the PVUR results, the highest VUF values were mainly observed near electrically remote feeder-end buses where voltage sensitivity is greater.

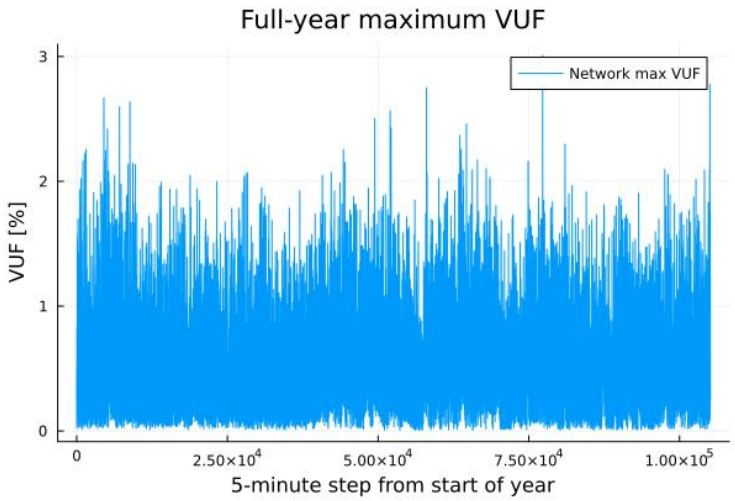


Figure 16: Full-year Maximum VUF

Compared with PVUR, VUF provides a more physically rigorous indicator of three-phase voltage imbalance because it directly considers symmetrical voltage components. The results confirm that uneven single-phase PV and load allocation can increase negative-sequence voltage levels in residential low-voltage feeders.

Overall, the VUF analysis further demonstrates that high DER penetration can significantly influence phase balance behaviour in practical residential distribution networks.

Metric	Value
Maximum VUF range across Task II cases	3.388% – 3.581%
Lowest maximum VUF	3.388% in Case 2 PV-only / PV-only SSO
Highest maximum VUF	3.581% in Case 3 SSO battery cases
Bus with highest VUF	b4a or b6a depending on case
VUF variation observed	Yes
Severity	Moderate

Table 7: VUF Summary

Across the Task II simulations, the maximum VUF ranged from approximately 3.39% to 3.58%. The PV-only cases had a maximum VUF of about 3.39%, while the standard battery cases produced a maximum VUF of about 3.52%. Under the SSO battery scenarios, the maximum VUF increased slightly to about 3.58%. This shows that battery operation did not uniformly reduce voltage unbalance. Since batteries were installed only at selected single-phase customers, their charging and discharging behaviour could change phase-specific power flows and increase the worst-case negative-sequence voltage component.

5 Critical Discussion

Although the rule-based HEMS developed in this study successfully coordinated rooftop PV generation, battery storage, and residential demand, several limitations were observed during the simulations.

First, the study adopted a net-load modelling approach in OpenDSS, where PV generation and battery operation were represented through equivalent net power injection at each residential load. This simplified the implementation of large-scale yearly simulations and reduced modelling complexity. However, this approach does not explicitly model inverter dynamics or detailed converter behaviour.

Second, the results demonstrate that increasing battery energy capacity alone does not completely eliminate reverse power flow or voltage rise. Although the 20 kWh battery case reduced annual feeder export more effectively than the 10 kWh case, the improvement remained relatively moderate because battery charging/discharging power was still limited to 5 kW. Therefore, battery power rating is also an important factor influencing feeder performance.

The SSO scenario also revealed several limitations of the rule-based controller. Under free-electricity periods, the controller allowed relatively aggressive battery charging from the grid. While this behaviour reduced customer electricity costs and increased battery utilisation, it also temporarily increased feeder import power. This demonstrates that local economic optimisation does not necessarily correspond to globally optimal feeder operation.

Another important observation is that the 10 kWh battery showed relatively limited improvement under the SSO scenario. Since the battery was already close to fully charged by midday rooftop PV surplus, the additional free-electricity period provided only limited additional charging opportunity. In contrast, the 20 kWh battery retained more available storage capacity and therefore exhibited more noticeable operational changes under SSO.

Overall, the study demonstrates that rule-based HEMS strategies are simple and computationally efficient, but they do not explicitly optimise feeder-level objectives such as voltage regulation, phase balancing, or network-wide power-flow minimisation. More advanced optimisation-based HEMS strategies could potentially achieve improved coordination between customer economics and network performance.

6 Conclusion

In this study, a rule-based Home Energy Management System (HEMS) was successfully developed and integrated into a low-voltage residential feeder model in OpenDSS. Building on the Task I network, rooftop PV systems, residential battery storage, and different electricity

tariff scenarios were investigated through both 24-hour Julia simulations and full-year quasi-static time-series (QSTS) simulations.

The results demonstrated that high rooftop PV penetration significantly changes feeder operating behaviour. Reverse power flow and daytime voltage rise became observable under PV-only conditions, particularly during periods of high solar generation.

After battery integration, reverse power flow and voltage rise were reduced due to increased local PV self-consumption. The 20 kWh battery case achieved further reduction in feeder export energy compared with the 10 kWh battery case; however, the improvement remained limited by the fixed 5 kW charging/discharging power constraint.

The SSO scenario further influenced battery dispatch behaviour by encouraging additional charging during the free-electricity period. Although this reduced reverse power flow and improved local battery utilisation, it also highlighted several limitations of the rule-based HEMS strategy, particularly the lack of feeder-level optimisation.

Voltage unbalance analysis using PVUR and VUF showed that uneven single-phase PV and load allocation can increase phase imbalance, especially near feeder-end buses. Nevertheless, all case studies remained within acceptable feeder voltage operating limits throughout the full-year simulations.

Overall, the study demonstrates that distributed rooftop PV and residential battery storage can significantly influence low-voltage feeder operation. While the rule-based HEMS provides a simple and effective control strategy, more advanced optimisation-based approaches could further improve both customer economic performance and network operation.

Case	Reverse Flow	Voltage Behaviour	Battery Effectiveness	Voltage Unbalance
Task 1	Not observed / low	Base-case voltage profile	-	Moderate
PV only	High	Voltage rise observed but within limit	-	Moderate
PV only + SSO	Same as PV only	Same as PV only	-	Same as PV only
10 kWh battery	Reduced	Within limit; max voltage slightly lower	Medium	Similar / slightly changed
20 kWh battery	Further reduced	Within limit; max voltage similar	Higher energy-shifting benefit	Similar to 10 kWh case
10 kWh battery + SSO	Further reduced	Within limit, not uniformly improved	Medium; affected by SSO charging	Higher worst-case PVUR/VUF

20 kWh battery + SSO	Lowest reverse-flow hours	Within limit, not uniformly improved	Highest export reduction	Highest worst-case PVUR/VUF

Table 8: Overall Simulation Summary

Github URL: https://github.sydney.edu.au/GregorVerbicGroup/ELEC5208_Task2_Group36

Task allocation:

Code: Zhengsheng Zhao

The Code Modification and Improvement: Haotian Guo

Results recording: Haotian Guo & Zhengsheng Zhao

Inspection of all work & Upload documents: Yiming Qin & Yang Jin

Report writing: Yiming Qin & Yang Jin